Section C Comprehensive Review - Mixed - Answer Key

1. $y = f(x) = (81)(3)^{x-4}$; $y = f(x) = (1)(3)^{x}$
2. y-intercept = 750; $r = 1.12$; 12% growth
3. $y = f(x) = (-10)\left(\frac{1}{4}\right)^{x}$
4. $y = f(x) = (-3)\left(\frac{2}{3}\right)^{x}$
5. $a_n = -25\left(-\frac{2}{5}\right)^{n-1}$
6. $a_1 = 11, \quad a_n = 4a_{n-1}$
7. $r = 4$; $a_n = 5(4)^{n-1}$; $a_6 = 5120$
8. $y = f(x) = (640)(0.75)^{x}$; at $x = 2$, $y = 360$
9. $y = f(x) = (8)(2)^{x-3}$; $y = f(x) = (1)(2)^{x}$; $y = f(x) = (4)(2)^{x-2}$
10. y-intercept = 900; $r = 0.65$; 35% decay
11. $y = f(x) = (8)(6)^{x}$
12. $a_n = -4(-3)^{n-1}$
13. $y = f(x) = (18)(3)^{x-1}$; $y = f(x) = (6)(3)^{x}$
14. $a_1 = 3, \quad a_n = -5a_{n-1}$
15. $y = f(x) = (200)(1.5)^{x}$; at $x = 3$, $y = 675$
16. $y = f(x) = (15)\left(\frac{4}{5}\right)^{x}$
17. $y = f(x) = (24)(2)^{x}$
18. $y = f(x) = (32)(2)^{x-2}$; $y = f(x) = (8)(2)^{x}$; $y = f(x) = (16)(2)^{x-1}$
19. $y = f(x) = \left(\frac{3}{2}\right)\left(\frac{1}{2}\right)^{x-2}$; $y = f(x) = (6)\left(\frac{1}{2}\right)^{x}$; $y = f(x) = (3)\left(\frac{1}{2}\right)^{x-1}$
20. $a_n = 8(-2)^{n-1}$
21. $a_n = 256\left(\frac{1}{2}\right)^{n-1}$
22. $r = 2$; $a_n = 7(2)^{n-1}$
23. $a_n = 10(3)^{n-1}$
24. $a_n = 144\left(\frac{1}{3}\right)^{n-1}$
25. $y = f(x) = (-12)(2)^{x-2}$; $y = f(x) = (-3)(2)^{x}$; $y = f(x) = (-6)(2)^{x-1}$
26. $r = \frac{1}{3}$; $a_n = 2700\left(\frac{1}{3}\right)^{n-1}$; $a_5 = \frac{100}{3}$
27. $a_n = 6(5)^{n-1}$
28. $y = f(x) = (5000)(0.9)^{x}$; at $x = 2$, $y = 4050$
29. $a_1 = 128, \quad a_n = \frac{1}{4}a_{n-1}$
30. $r = -2$; $a_n = 5(-2)^{n-1}$
31. $a_n = 192\left(\frac{1}{2}\right)^{n-1}$
32. $a_1 = -18, \quad a_n = \frac{2}{3}a_{n-1}$